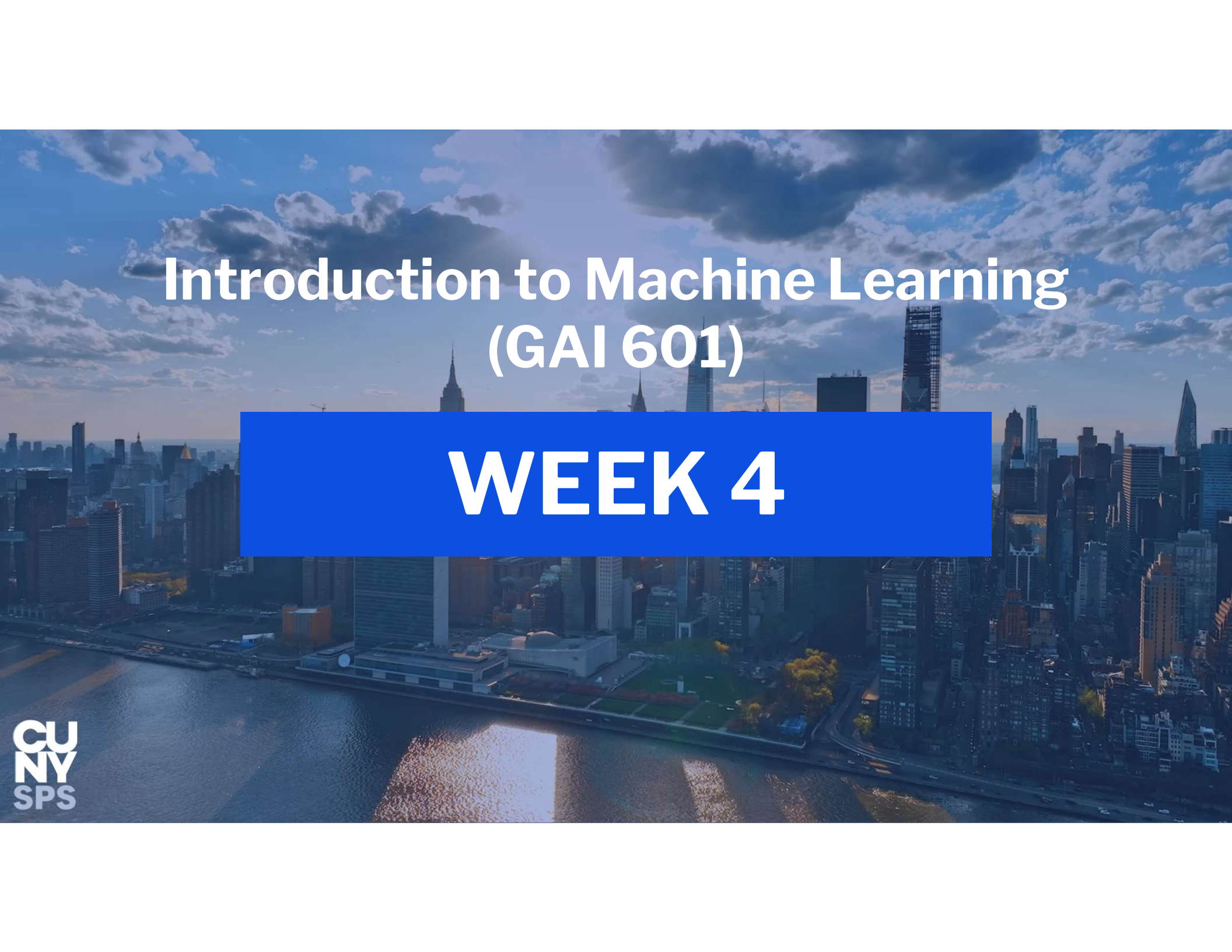
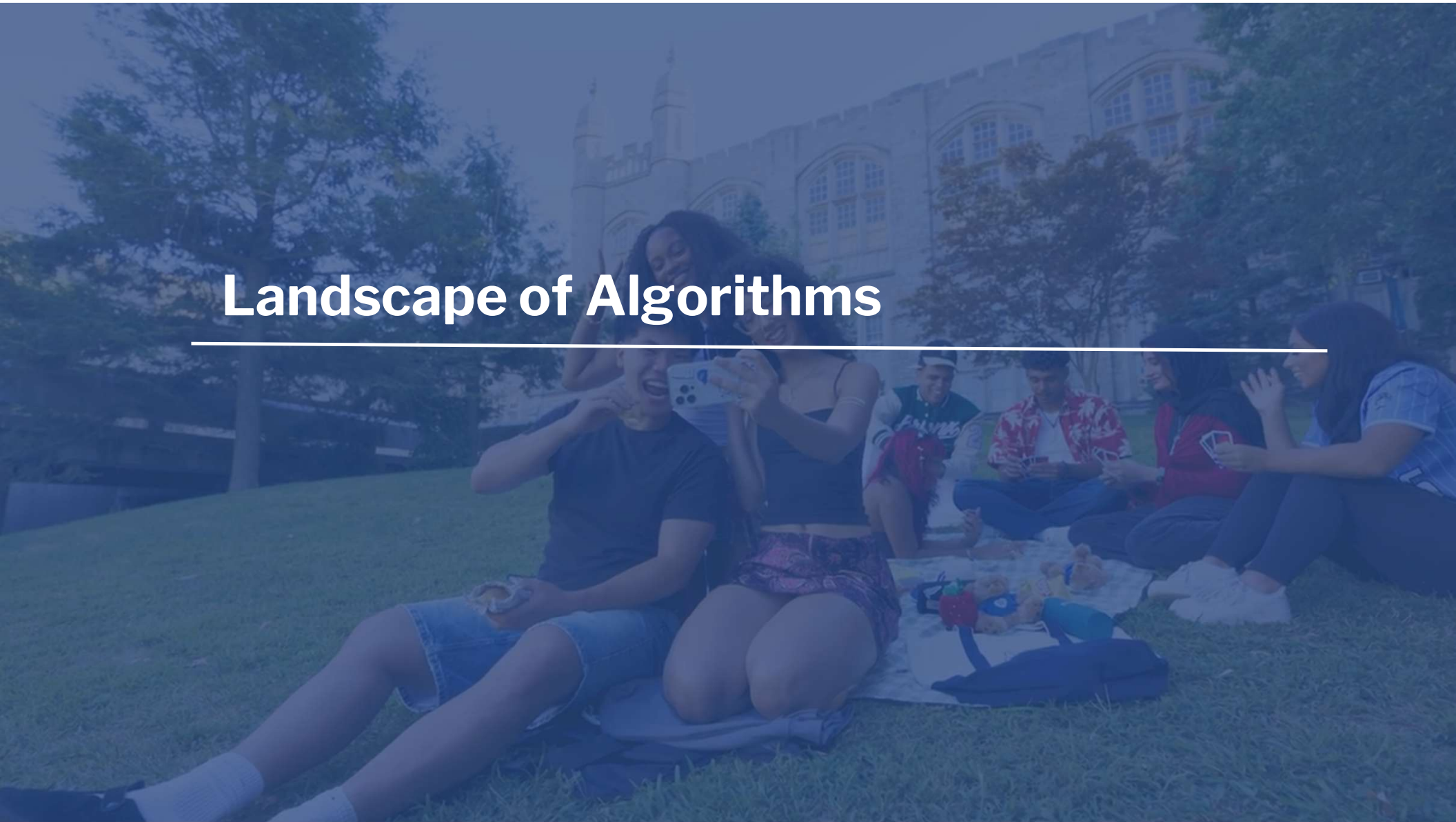


An aerial photograph of the New York City skyline, featuring numerous skyscrapers and the Hudson River in the foreground. A large blue rectangular box is centered over the image, containing white text. The sky is blue with scattered white clouds.

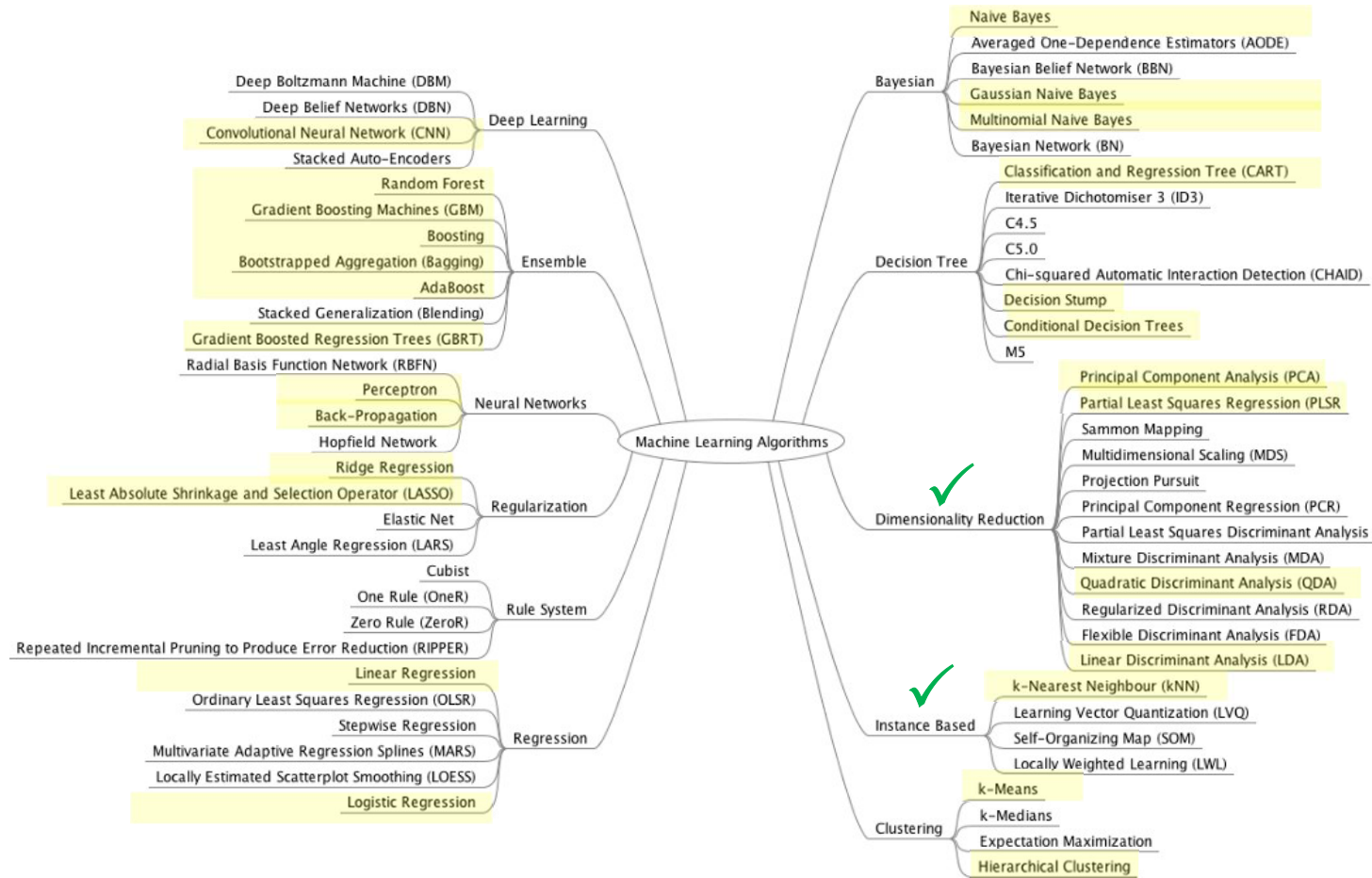
Introduction to Machine Learning (GAI 601)

WEEK 4

Landscape of Algorithms

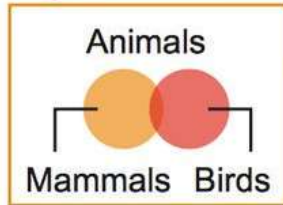


Landscape of Algorithms



5 Tribes onf Machine Learning

Symbolists



Use symbols, rules, and logic to represent knowledge and draw logical inference

Favored algorithm

Rules and decision trees

Bayesians

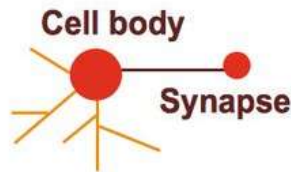


Assess the likelihood of occurrence for probabilistic inference

Favored algorithm

Naive Bayes or Markov

Connectionists



Recognize and generalize patterns dynamically with matrices of probabilistic, weighted neurons

Favored algorithm

Neural networks

Evolutionaries

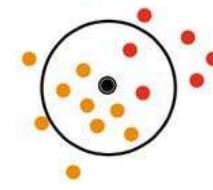


Generate variations and then assess the fitness of each for a given purpose

Favored algorithm

Genetic programs

Analogizers



Optimize a function in light of constraints ("going as high as you can while staying on the road")

Favored algorithm

Support vectors

Source: Pedro Domingos, *The Master Algorithm*, 2015

Naïve Bayes



Bayes Theorem

LIKELIHOOD

The probability of "B" being True, given "A" is True

PRIOR

The probability "A" being True. This is the knowledge.

$$P(A|B) = \frac{P(B|A).P(A)}{P(B)}$$

The diagram shows the formula for Bayes' Theorem. A yellow arrow points from the 'LIKELIHOOD' text to the $P(B|A)$ term in the numerator. Another yellow arrow points from the 'PRIOR' text to the $P(A)$ term in the numerator. A third yellow arrow points from the 'POSTERIOR' text to the $P(A|B)$ term on the left side of the equation. A fourth yellow arrow points from the 'MARGINALIZATION' text to the $P(B)$ term in the denominator.

POSTERIOR

The probability of "A" being True, given "B" is True

MARGINALIZATION

The probability "B" being True.

Naïve Bayes: An Example

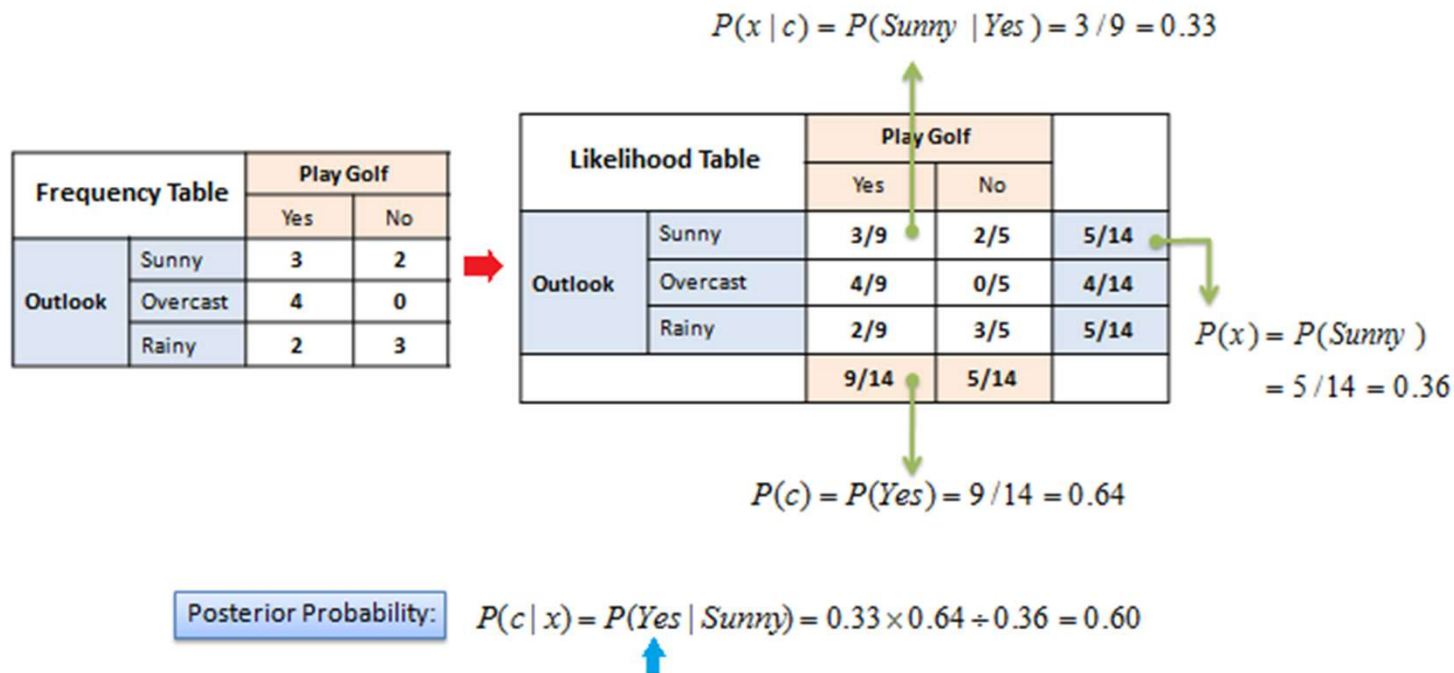
Predicting whether you should play golf

Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

Source: Supervised Learning Algorithms: A Comparison, November 2020Kristu Jayanti Journal of Computational Sciences (KJCS) - DOI:10.59176/kjcs.v1i1.1259. Free access

Naïve Bayes: An Example

Let's look at the data:



Naïve Bayes: An Example

Frequency Table

		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3



Likelihood Table

		Play Golf	
		Yes	No
Outlook	Sunny	3/9	2/5
	Overcast	4/9	0/5
	Rainy	2/9	3/5

		Play Golf	
		Yes	No
Humidity	High	3	4
	Normal	6	1



		Play Golf	
		Yes	No
Humidity	High	3/9	4/5
	Normal	6/9	1/5

		Play Golf	
		Yes	No
Temp.	Hot	2	2
	Mild	4	2
	Cool	3	1



		Play Golf	
		Yes	No
Temp.	Hot	2/9	2/5
	Mild	4/9	2/5
	Cool	3/9	1/5

		Play Golf	
		Yes	No
Windy	False	6	2
	True	3	3



		Play Golf	
		Yes	No
Windy	False	6/9	2/5
	True	3/9	3/5

Naïve Bayes: An Example

Will I play golf in the following example?

Outlook	Temp	Humidity	Windy	Play
Rainy	Cool	High	True	?

$$P(Yes | X) = P(Rainy | Yes) \times P(Cool | Yes) \times P(High | Yes) \times P(True | Yes) \times P(Yes)$$

$$P(Yes | X) = 2/9 \times 3/9 \times 3/9 \times 3/9 \times 9/14 = 0.00529 \rightarrow 0.2 = \frac{0.00529}{0.02057 + 0.00529}$$

$$P(No | X) = P(Rainy | No) \times P(Cool | No) \times P(High | No) \times P(True | No) \times P(No)$$

$$P(No | X) = 3/5 \times 1/5 \times 4/5 \times 3/5 \times 5/14 = 0.02057 \rightarrow 0.8 = \frac{0.02057}{0.02057 + 0.00529}$$

Naïve Bayes: Use-cases

- **Spam detection**
- **Sentiment analysis (news articles)**
- **Document classification**
- **Many classification problems...**

Source: Supervised Learning Algorithms: A Comparison, November 2020 Kristu Jayanti Journal of Computational Sciences (KJCS) - DOI:10.59176/kjcs.v1i1.1259. Free access

Naïve Bayes: Strengths and Weaknesses

Strengths:

- Simplicity and computational efficiency.
- It does a great job handling categorical features directly, without any preprocessing.
- Outperforms more sophisticated classifiers when working with a large number of predictors
- It handles noisy and missing data pretty well.

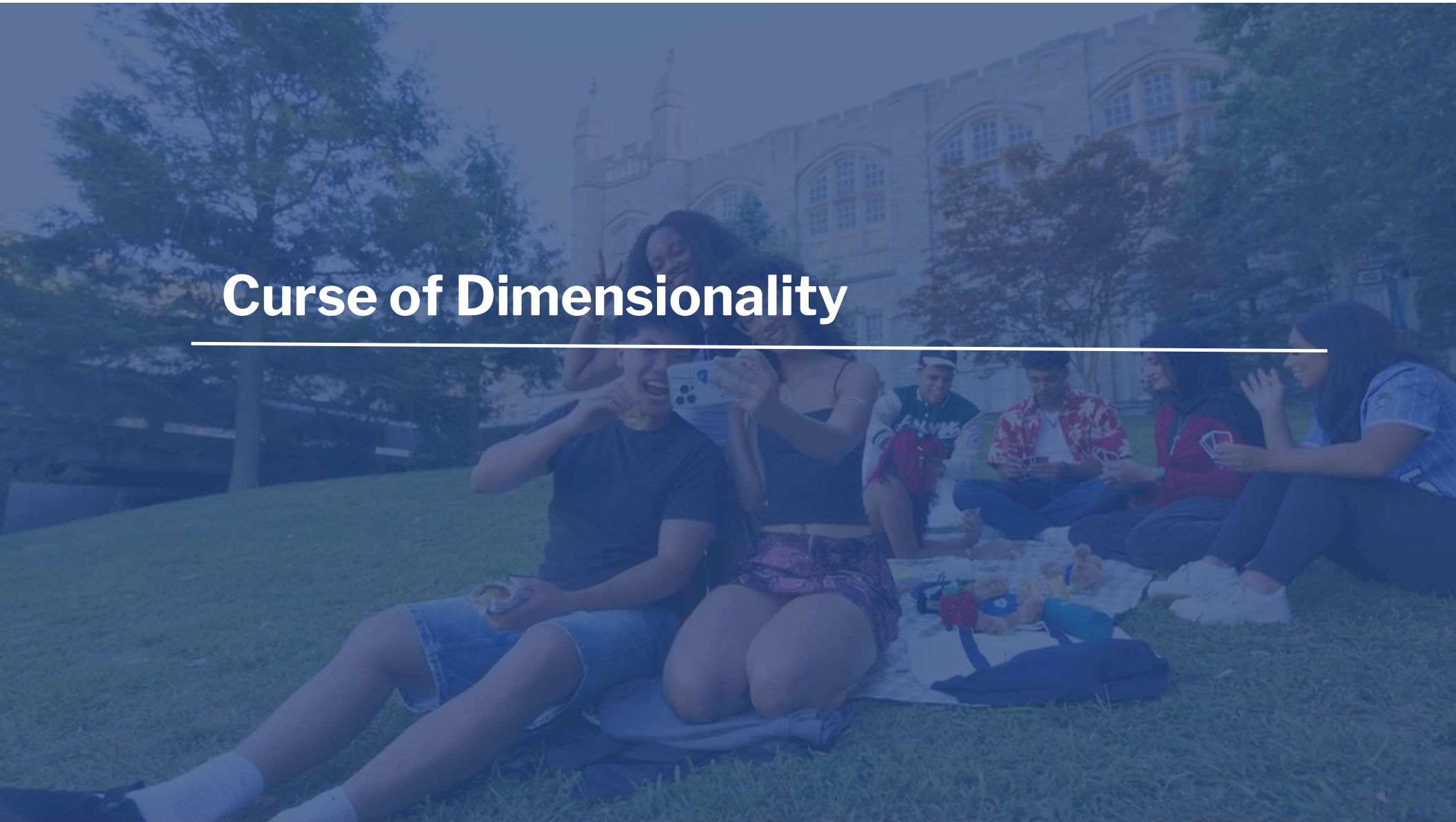
Weaknesses:

- Needs a sizable amount of data
- It is naïve: assumption of independence between inputs & classes
- Doesn't work well for datasets with a large number of continuous features
- It assumes that all features within a class are not only independent but are equally important

Naïve Bayes: Use-cases

- **Spam detection**
- **Sentiment analysis**
- **Document classification**
- **Recommendation System**
- **Medical Diagnosis**
- **Weather Prediction**
- **Face Recognition**
- **Many classification problems...**

Curse of Dimensionality



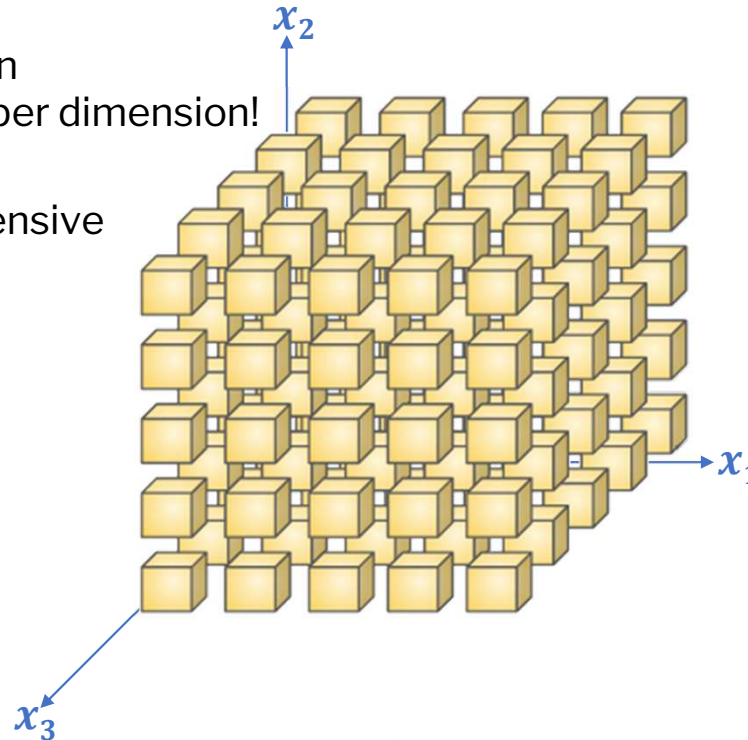
Curse of Dimensionality

- **Data is sparse**

The Iris data set has 150 instances in 4-dimensions: average ~3.5 values per dimension!

- **Data is scarce**

Labeled data is hard to get and expensive (est. \$2/instance on average for outsourced labeling services)

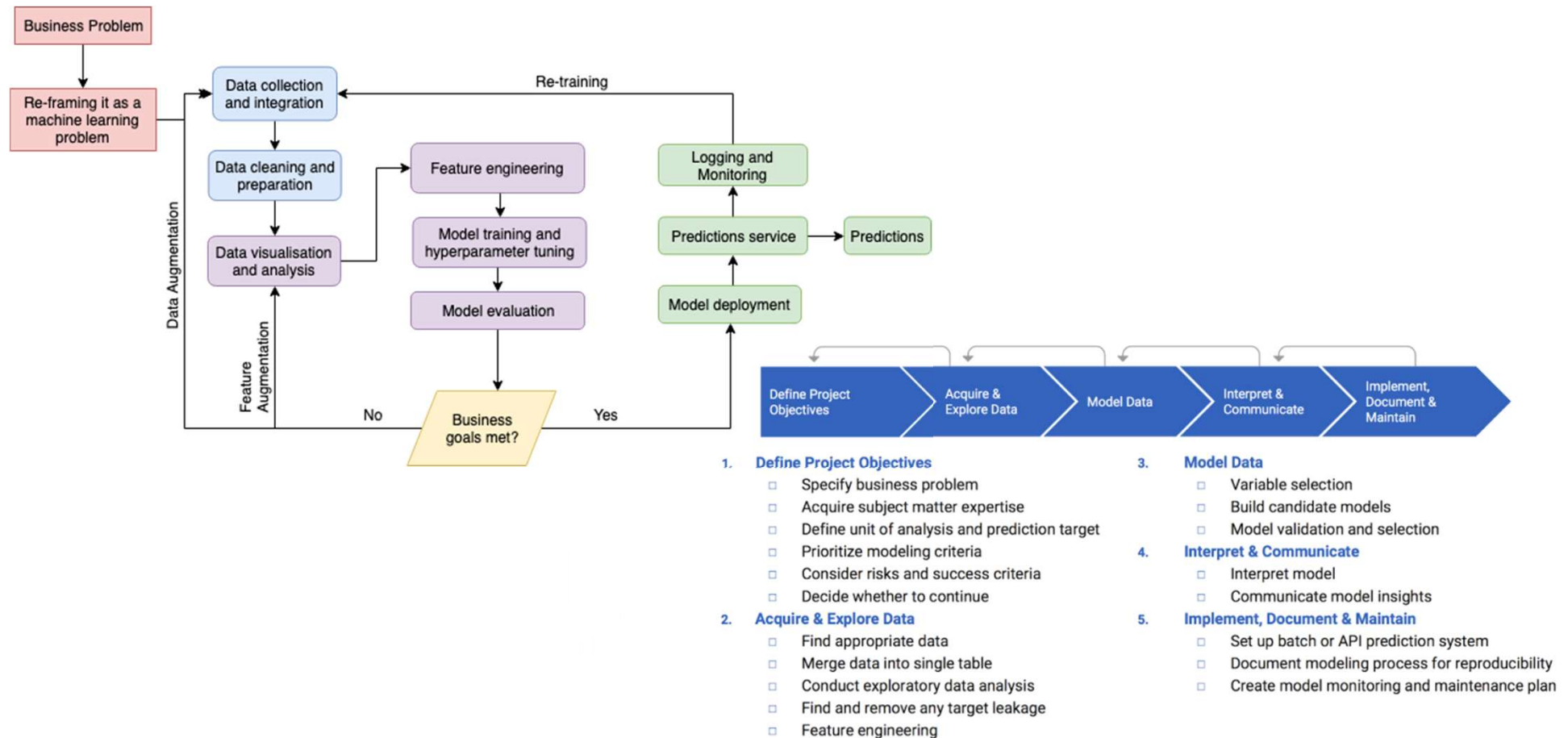


Source: therbootcamp.github.io

A group of students are sitting on a green lawn in front of a large, historic stone building with multiple windows and a tower. The students are engaged in various activities, including taking photos with their phones and playing cards. The scene is overlaid with a semi-transparent blue filter. The title 'Assignment 1: Defining Business Objectives' is written in white, bold, sans-serif font across the middle of the image, underlined.

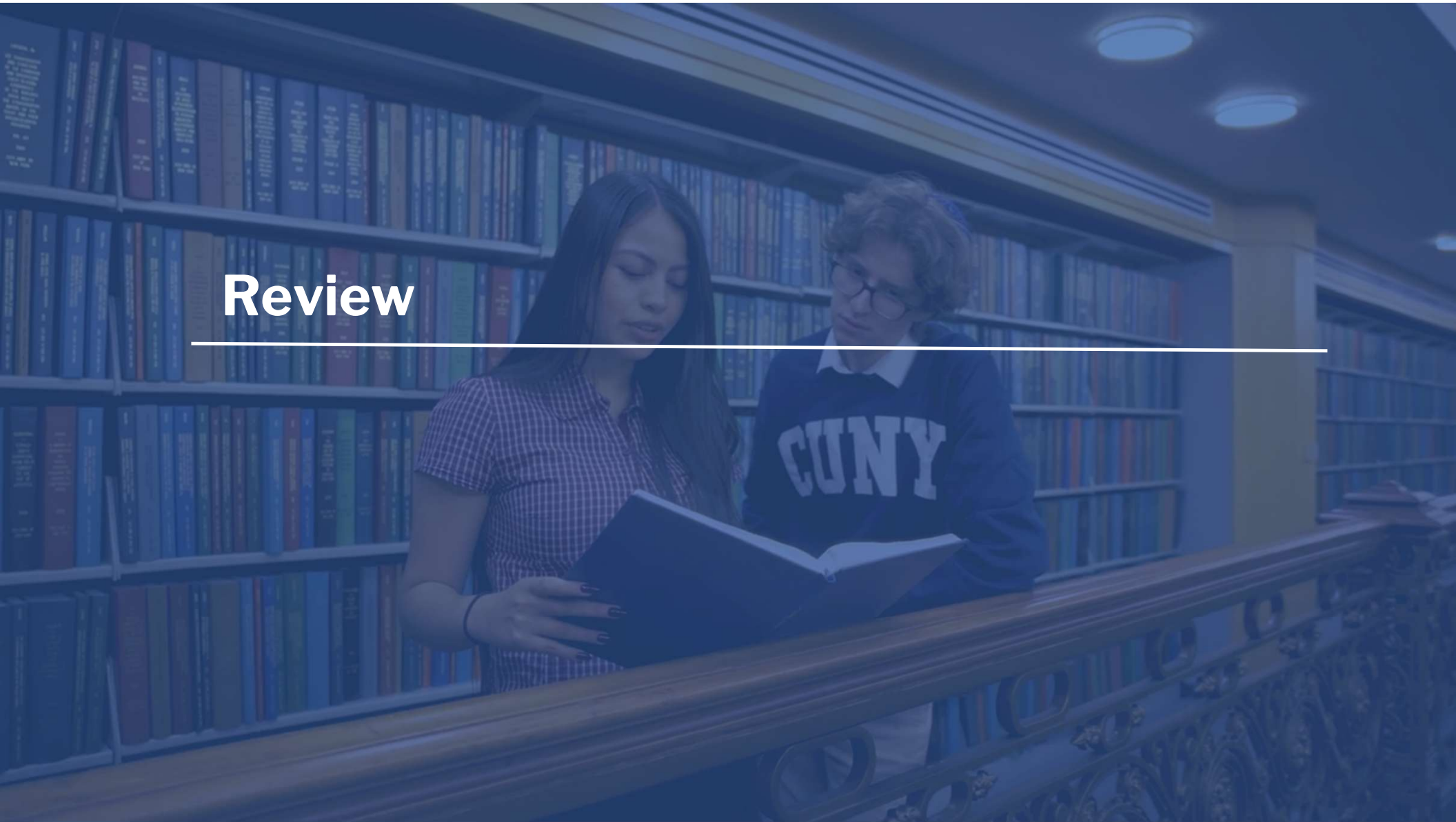
Assignment 1: Defining Business Objectives

Defining Business Objectives



Source: Datadog & Domino

Review



This week we covered

Lesson Objectives/Topics

1. Evaluate a Naive Bayes classifier using categorical and numeric data.
2. Apply Bayes' theorem to interpret classification probabilities.
3. Preprocess text data for input into Naive Bayes models.
4. Integrate multiple classification methods to support decision-making
5. The Curse of Dimensionality and sparseness of data

An aerial photograph of the Manhattan skyline, featuring the Manhattan Bridge in the foreground and the dense cityscape in the background. The bridge's stone towers and suspension cables are prominent. The water of the Hudson River is visible in the foreground. The sky is clear and blue.

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